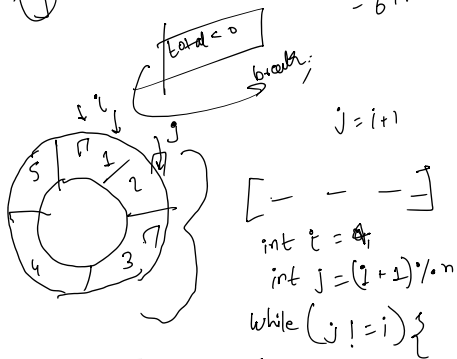


gas = [1, 2, 3, 4, 5]
cost = [3, 4, 5, 1, 2]

$$\text{total} = 6 + 1 - 3 = 4$$

$$3 + 5 = 8 - 2 = 6 + 1$$



gas = [1, 2, 3, 4, 5], cost = [3, 4, 5, 1, 2]

```
for (int i = 0; i < n; i++) {
    if (gas[i] - cost[i] < 0) continue;
    int j = (i+1) % n;
    int cost_of_going = cost[i];
    int gas_gained = gas[j];
    int total = gas[i] - cost_of_going + gas_gained;
    while (j != i) {
        if (total < cost[j]) break;
        int cost_of_going = cost[j];
        j = (j+1) % n;
        int new_gas_gained = gas[j];
        total = total - cost_of_going + new_gas_gained;
    }
    if (i == j) return i;
}
return -1;
```

[1, 2, 3, 4, 5]

[3, 4, 5, 1, 2]

$$\text{gas} - \text{cost} < 0$$

[1, 2, 3, 4, 5]

[3, 4, 5, 1, 2]

[-ve) (+ve) (-ve) (+ve) (+ve)